

AUTHOR SELF-ASSESSMENT

# Architecture Integrity and NIST Readiness Review

*Publication integrity, technical coherence, mapping readiness, validation gaps, and engagement gates*

| Field               | Controlled value                                       |
|---------------------|--------------------------------------------------------|
| Document identifier | AERA-001-I1-2026-08-07                                 |
| Package build       | AERA-001-1.2WD-20260807.1                              |
| Subject             | AERA 1.2 WD controlled publication package             |
| Prepared by         | Angelis Pseftis                                        |
| Affiliation         | Independent author                                     |
| Assessment date     | 7 August 2026                                          |
| Assurance level     | I1 author self-assessment; not independent             |
| Review status       | Developmental-review candidate; public review not open |

**STATUS:** This report records an author-conducted I1 review of a working draft. It is not an independent audit, NIST assessment, certification, consensus determination, or government endorsement.

## Executive Disposition

**AERA 1.2 WD is a developmental-review candidate suitable for structured NIST-facing engagement preparation; public review is not open, and the package is not ready for a NIST support, endorsement, adoption, conformance, or OLIR-publication claim.**

**DISPOSITION:** Proceed to release readiness only after final package hash, metadata, rights, and publication-authorization checks. Public review remains closed until those gates are formally cleared. Hold all stronger claims until independent review, consensus controls, atomic mapping adjudication, and multi-organization validation are complete.

AERA addresses a genuine architecture boundary: resource access, identity, model quality, and policy statements do not alone establish authority to cause a specific consequence. The framework's strongest contribution is its treatment of the complete delegated-authority transaction - from grant through observed effect, recovery, and closure - as the unit of assurance.

| Review area                      | I1 result                | Evidence basis / limitation                                                                                 |
|----------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|
| Controlled requirements          | PASS                     | 199 unique 1.2 WD rows in the generated canonical register.                                                 |
| Threat traceability              | PASS                     | 25 decision-authority threat rows in the generated matrix.                                                  |
| Controlled sources               | PASS WITH STATUS CONTROL | 69 source rows; draft, living, and final status must remain distinct.                                       |
| Normative boundary               | PASS AT SOURCE           | Normative modalities, canonical register, conformance states, and publication-defect rule are explicit.     |
| NIST mapping                     | DEVELOPMENTAL            | Candidate lanes exist; atomic target identifiers, rationales, and independent SME adjudication remain open. |
| Consensus and empirical validity | OPEN                     | No completed balanced consensus cycle, multi-organization pilot, or inter-rater study.                      |
| Release integrity                | CONDITIONAL              | Final signed/hash manifest, license terms, and complete fixed-layout package remain release gates.          |

**CLAIM BOUNDARY:** NIST compatibility is a design objective. A NIST citation, community participation, mapping submission, or comment acknowledgment would not establish NIST support for AERA.

# 1. Review Scope, Method, and Assurance Boundary

## 1.1 Manifest-scoped review

This assessment covers the AERA 1.2 WD controlled source state and the document, register, threat, source, comment, and mapping artifacts generated from it. It does not extend to deployment performance, legal sufficiency, sector authorization, or a third-party implementation. Final hash verification is intentionally deferred until every release artifact is frozen.

| Assurance tier | Meaning                                                                     | Status for this report                                            |
|----------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------|
| I0             | Automated build, structural, formula, metadata, and render checks.          | Applied where tooling exists; package-level freeze checks remain. |
| I1             | Author or maintainer reviews content, mappings, and release integrity.      | This report.                                                      |
| I2             | Independent technical and publication review by qualified reviewers.        | Not completed.                                                    |
| I3             | Multi-organization implementation, inter-rater, and operational validation. | Not completed.                                                    |

**IMPORTANT:** I0-I3 are AERA review-planning labels, not NIST assurance levels and not an external certification scheme.

## 1.2 Review activities

- Reconciled the generated requirement, threat, and source counts and examined the specification source for status, authority, conformance, mapping, governance, and evidence-boundary controls.
- Assessed NIST-facing claims against AI RMF, CSF, SP 800-53, IR 8477, National OLIR, AIRC, and current engagement pathways.
- Separated completed publication corrections from open empirical, consensus, licensing, and external-review work; defined objective release blockers and closure evidence.

## 1.3 Limitations

This review uses public publisher-controlled source information. It does not assert clause-level equivalence to proprietary ISO or IEEE standards without authorized full-text review. Continuously maintained sources require version or access-date control at each release.

## 2. Publication and Cross-Document Integrity

The 1.2 WD package corrects the central publication-control failures observed in the 1.1 RC1 set: stale fixed-layout navigation, source drift between human- and machine-readable artifacts, references to filenames not actually released, and an improperly characterized same-author 'independent' review.

| Integrity object          | Controlled 1.2 WD state                                                 | Release condition                                                                        |
|---------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Edition and status        | AERA-001, Version 1.2 WD; independently authored working draft.         | Every artifact and metadata field must use the same edition/status.                      |
| Normative requirements    | 199 active rows; Appendix A / controlled registry is canonical.         | DOCX, PDF, XLSX, and CSV text must reconcile exactly.                                    |
| Threat-control-test trace | 25 DAT rows.                                                            | Every referenced AERA requirement must resolve; all test/evidence fields remain present. |
| Controlled sources        | 69 rows.                                                                | Edition, publication status, access date, direction, and caveat must reconcile.          |
| Review records            | Immutable reviewer-entered comment identifiers in the 1.2 matrix.       | No row-derived identifier may change when rows are sorted or inserted.                   |
| Machine mappings          | Separate developmental ledger for AI RMF, CSF 2.0, and SP 800-53 lanes. | No relationship claim without one atomic pair and adjudication metadata.                 |
| Package inventory         | Actual 1.2 filenames only.                                              | Final checksum manifest must match every published byte stream.                          |

### 2.1 Normative-control improvements

- Conformance outcomes are separated: CONFORMING, CONDITIONALLY CONFORMING, NONCONFORMING, and INDETERMINATE.
- The normative perimeter identifies Section 1, Section 3 definitions, uniquely identified requirements, and clauses or tables explicitly marked [NORMATIVE].
- Different text for one requirement identifier is a release-blocking publication defect; it does not create an alternate requirement.
- Applicability and assessment result are separate, with evidence, owner, inheritance, deviation, approval, and invalidating-condition fields.
- Capabilities are reported by evidenced operational state: PRESENT, MONITORED, ACTIVE IN PATH, USED IN COMPUTATION, or VALIDATED EFFECTIVE.

**RESIDUAL CONTROL:** The published PDF must be frozen, signed or hashed, and reconciled to the editable and machine-readable derivatives before any release-candidate integrity claim.

### 3. Technical Architecture Assessment

**ARCHITECTURAL JUDGMENT:** AERA is technically differentiated where it binds delegated authority, current evidence, policy, consequence, time/PNT, independent enforcement, effect observation, and recovery into one reconstructable transaction.

| Architecture strength            | Assessment                                                                           | Evidence still required                                                               |
|----------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Authority as protected object    | Stronger than treating identity or resource access as sufficient authorization.      | Reference policy semantics; attenuation and revocation interoperability tests.        |
| Proposal-to-effect lifecycle     | Captures attempted effect, achieved effect, divergence, recovery, and closure.       | Event schema; distributed ordering rules; effect-observer qualification.              |
| Visible assurance vector         | Avoids collapsing distinct claims into one confidence score.                         | Calibration, precedence, uncertainty, and consequence-specific acceptance criteria.   |
| Consequence-based control floors | Supports proportional human control, evidence independence, and recovery.            | Sector hazard analysis; validated classification guidance; disagreement studies.      |
| Continuous reassessment          | Correctly invalidates stale authority, evidence, threat, timing, or integrity state. | Latency budgets, fail-safe semantics, race/partition tests, and common-mode analysis. |
| Technology neutrality            | Keeps AI, cloud, quantum, sensing, OT, and PNT subordinate to governed authority.    | Representative implementation patterns and interoperability profiles.                 |

#### 3.1 Known semantic risk

Disposition vocabulary and legacy requirement lineage require strict control. The canonical set should use PERMIT, PERMIT-WITH-CONSTRAINTS, DEFER, ESCALATE, SIMULATE, QUARANTINE, and DENY consistently. Any change to an existing requirement identifier must be recorded as a versioned normative revision, not silently normalized in a support document.

#### 3.2 Evidence boundary

Document consistency demonstrates publication quality, not operational effectiveness. AERA's maturity and conformance methods remain proposed constructs until representative systems produce repeatable evidence under independent assessment.

## 4. NIST Compatibility Assessment

AERA is best positioned as an implementation-oriented reference architecture that can support organizations using NIST risk outcomes. It should not present itself as an alternate NIST framework, a replacement for existing controls, or a pre-adjudicated mapping.

| NIST anchor             | AERA contribution                                                                                                  | Permitted claim at 1.2 WD                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| AI RMF 1.0              | Decision-authority boundary, runtime evidence, enforcement, observed effect, recovery, and accountable delegation. | Designed for compatibility with AI RMF outcomes; detailed mapping remains developmental. |
| CSF 2.0                 | Governance-to-operation trace across identity, protection, detection, response, recovery, and decision effects.    | Directional comparison only; no CSF conformance claim.                                   |
| SP 800-53 / 53A         | Candidate control and assessment evidence reuse at an explicitly bounded implementation and evidence period.       | Candidate atomic mappings; implementation and effectiveness not inferred.                |
| SP 800-207              | Access is necessary but insufficient; AERA adds bounded decision and effect authority.                             | Complementary semantic relationship, not equivalence.                                    |
| IR 8323 Rev. 1          | Decision-specific time/PNT source state, uncertainty, integrity, use, and degraded behavior.                       | PNT profile comparison; draft successors remain change triggers.                         |
| IR 8477 / National OLIR | A controlled path for one-to-one concept relationships and machine-readable mapping metadata.                      | Mapping-development ledger only; not OLIR submitted, published, or reviewed.             |
| OSCAL                   | Candidate future representation for AERA requirements, profiles, assessment results, and evidence links.           | Interoperability design objective; no OSCAL artifact claim yet.                          |

### 4.1 AI RMF trustworthiness coverage

The working draft now explicitly evaluates validity and reliability, safety, security and resilience, accountability, transparency, explainability, interpretability, privacy enhancement, and management of harmful bias. Coverage in prose or requirements is not proof of outcome effectiveness; metrics, test methods, and pilot evidence remain necessary.

**READINESS JUDGMENT:** The architecture is NIST-compatible in intent and terminology control, but not NIST-validated. That distinction must remain visible on the cover, in release metadata, in mapping ledgers, and in all public statements.

## 5. IR 8477 and OLIR Mapping Readiness

The legacy D/P/E/C/N codes are useful working annotations but are not mechanically translatable to NIST IR 8477 relationship semantics. Compound entries must be decomposed into atomic source-target assertions.

| Required mapping field           | 1.2 WD state                                                 | Closure evidence                                                                           |
|----------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Focal document element           | AERA requirement identifiers are stable in the 1.2 register. | Exact normative text and edition checksum.                                                 |
| Reference document element       | Candidate standards lanes exist.                             | Exact target identifier, title/text, edition, and authoritative source.                    |
| Relationship style and direction | IR 8477 method described.                                    | One allowed relationship per atomic pair; no inferred conversion from legacy codes.        |
| Rationale                        | Development field available.                                 | Element-specific semantic rationale covering scope, modality, conditions, and outcome.     |
| Review status                    | Developmental.                                               | Named mapper and independent SME reviewer, date, decision, dissent, and disposition.       |
| Submission metadata              | Not complete.                                                | OLIR template validation, license/terms, point of contact, version, and public access URI. |

### 5.1 Minimum adjudication rule

1. Split every compound AERA-to-source mapping into one AERA element and one external element.
2. Quote or cite the exact authoritative target element without exceeding applicable license permissions.
3. Select the IR 8477 relationship only after scope, modality, preconditions, and expected outcomes are compared.
4. Record a concise rationale, editions, status, reviewer, dissent, and resolution.
5. Validate the machine-readable template and public licensing boundary before submission.

**PROHIBITED CLAIM:** OLIR-ready, submitted, accepted, cataloged, or NIST-endorsed are distinct states. None may be claimed unless independently evidenced for the exact published mapping artifact.

## 6. Open Development, Consensus, and Governance Readiness

NIST-compatible technical content is necessary but insufficient for institutional credibility. AERA needs a transparent maintenance and evidence process that separates editorial authority from external review, pilot evidence, assessment, and adoption decisions.

| Governance capability  | Working-draft treatment                                                                              | Open evidence                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Public access          | Developmental review package and comment matrix are defined.                                         | Stable public repository/URI and declared redistribution license.                         |
| Role separation        | Editor, contributor, reviewer, pilot, and assessor roles are identified.                             | Named independent reviewers; conflict and sponsorship disclosures.                        |
| Balanced participation | Multistakeholder review is required before consensus claims.                                         | Published stakeholder model, outreach record, and participation analysis.                 |
| Comment disposition    | Substantive dissent and versioned dispositions are required.                                         | Public issue ledger with accepted, modified, deferred, rejected, and withdrawn decisions. |
| Change control         | Impact analysis spans definitions, requirements, interfaces, threats, schemas, mappings, and claims. | Release board record, migration notes, and semantic diff.                                 |
| Maintenance            | Stable identifiers and non-silent version migration are required.                                    | Published cadence, responsible body, escalation path, and security advisory process.      |

### 6.1 Publication architecture recommendation

Before a consensus-oriented release, separate an outcome-focused voluntary AERA Core from the normative reference architecture, sector profiles, and conformance-assessment procedures. Preserve stable identifiers and traceability. Use of the outcome Core alone must never be represented as implementation of the architecture or satisfaction of a profile.

**LICENSE GATE:** Public redistribution, contribution acceptance, derivative works, patent/rights representations, and standards submissions require explicit license and contribution terms approved by the author.



## 7. Empirical Validation and Assessment Readiness

AERA needs evidence that the architecture is implementable, the requirements are interpreted consistently, controls work in representative conditions, and the maturity/conformance methods produce reproducible results.

| Validation objective      | Minimum method                                                                                                  | Acceptance evidence                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Implementability          | Two or more heterogeneous reference implementations across different decision classes.                          | Schemas, traces, latency/failure data, deviations, and implementation lessons.          |
| Threat coverage           | Exercise all 25 DAT rows under applicable nominal, adversarial, degraded, partitioned, and recovery conditions. | Negative-test results, residual threats, and evidence-quality assessment.               |
| Conformance repeatability | Independent assessors evaluate the same bounded evidence package.                                               | Agreement statistics, disagreement taxonomy, adjudication record, and revised guidance. |
| Maturity validity         | Compare assigned levels with observed operating behavior over time.                                             | Predictive/construct evidence, adverse cases, overrides, and level-change sensitivity.  |
| Human control             | Representative operators exercise oversight, takeover, escalation, and recovery by consequence class.           | Timing, workload, error, near-miss, usability, and mission-effect data.                 |
| Cross-domain resilience   | Partition, stale-state, common-mode, supply-chain, PNT, and enforcement-outage scenarios.                       | Demonstrated safe degradation, inhibition, reconciliation, and recovery bounds.         |

### 7.1 Proposed pilot sequence

- Pilot 1: bounded CC-0/CC-1 shadow and reversible decision classes; validate transaction observability and evidence capture.
- Pilot 2: CC-2 material actions with independent enforcement, exercised recovery, and explicit cumulative-risk limits.
- Pilot 3: cross-domain or degraded-operation cases with separate operators and assessors.
- Only after successful lower-consequence work should CC-3/CC-4 claims be considered, subject to sector-specific safety, legal, and authorization regimes.

**NO INFERENCE:** A successful pilot validates only its declared subject, decision population, configuration, context, evidence period, and acceptance criteria. It does not validate the framework universally.

## 8. Findings, Release Gates, and Closure Evidence

| ID    | Priority | Finding                                                                | Required closure evidence                                                                                    |
|-------|----------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| I1-01 | P0       | Redistribution and contribution terms are not yet declared.            | Approved license, contribution terms, rights statement, and publication authorization.                       |
| I1-02 | P0       | Final fixed-layout release and byte-level manifest are not yet frozen. | Rendered master/support PDFs, metadata validation, complete SHA-256 manifest, and clean-room file inventory. |
| I1-03 | P1       | IR 8477/OLIR mapping assertions are developmental.                     | Atomic mappings with exact targets, rationale, independent SME disposition, and template validation.         |
| I1-04 | P1       | No independent I2 technical/publication review has been completed.     | Qualified reviewer records, conflicts, defect dispositions, and closure verification.                        |
| I1-05 | P1       | No consensus process has been completed.                               | Defined participation model, review period, public comments, preserved dissent, and published dispositions.  |
| I1-06 | P1       | Conformance and maturity models lack I3 empirical validation.          | Multi-organization pilots, inter-rater study, metrics, limitations, and adverse results.                     |
| I1-07 | P1       | Reference interfaces and event schemas remain design work.             | Versioned schemas, conformance tests, compatibility policy, and implementation evidence.                     |
| I1-08 | P2       | Usability and assessor guidance require external testing.              | Task-based evaluation, reviewer feedback, ambiguity log, and revised guidance.                               |

**RELEASE DECISION:** AERA 1.2 WD may be published only as a developmental external technical discussion draft after P0 release controls close. P1 and P2 items may remain open if they are visible, owned, time-bounded, and do not support stronger claims.

## 9. Recommended NIST Engagement Path

The strongest near-term path is evidence-first participation, not a request for endorsement. Each engagement should contribute bounded technical artifacts that can stand independently of AERA branding.

1. Participate in the Trustworthy AI in Critical Infrastructure Profile community with concrete authority, guardrail, TEVV, recovery, and IT/OT mission-thread examples.
2. Use the AI Risk Management Framework community resources to test whether AERA outcomes, metrics, and evidence improve practical risk management.
3. Develop IR 8477-conformant atomic mapping lanes and seek independent subject-matter adjudication before considering a National OLIR submission.
4. Consider an organization-level AI Consortium contribution only after license, contribution, affiliation, sponsorship, and conflict terms are resolved.
5. Track the AI Agent Standards Initiative and related NCCoE work as context; respond to formal opportunities with bounded technical evidence.

### 9.1 Engagement language

| Use                                                    | Do not use                                                           |
|--------------------------------------------------------|----------------------------------------------------------------------|
| Independently authored AERA 1.2 Working Draft          | NIST-supported, NIST-approved, or NIST-endorsed                      |
| Designed for compatibility with selected NIST outcomes | NIST-aligned without a defined mapping scope and adjudication status |
| Candidate mapping under IR 8477 methodology            | NIST mapping, OLIR-ready, or OLIR-published without evidence         |
| Contributing technical input to an open process        | NIST is adopting AERA                                                |
| Pilot evidence for a bounded implementation            | AERA is validated or certified                                       |

## 10. Selected Authoritative Sources

These sources define the NIST baselines, mapping method, and current participation pathways referenced by this I1 self-assessment.

NIST AI 100-1. Artificial Intelligence Risk Management Framework (AI RMF 1.0). [Open official source for NIST AI 100-1](#)

NIST AI RMF. Program and revision status page. [Open official source for NIST AI RMF](#)

NIST IR 8477. Mapping Relationships Between Documentary Standards, Regulations, Frameworks, and Guidelines. [Open official source for NIST IR 8477](#)

National OLIR. Online Informative References Program. [Open official source for National OLIR](#)

NIST CI Profile COI. Trustworthy AI in Critical Infrastructure profile concept and community. [Open official source for NIST CI Profile COI](#)

NIST AI Consortium. Letter of interest and participation process. [Open official source for NIST AI Consortium](#)

NIST AI Agent Standards. AI Agent Standards Initiative. [Open official source for NIST AI Agent Standards](#)

**FINAL I1 VERDICT:** The 1.2 WD package is materially stronger and appropriate for bounded technical review. NIST compatibility is plausible and worth pursuing through open technical engagement, but NIST support cannot be promised, inferred, or represented without explicit NIST action and authorization.